Time Horizon Weather Prediction Machine Learning Model

A Comparative Analysis of ML Techniques for Multi-Step Forecasting

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The Evolution of Weather Forecasting

Traditional NWP

Numerical Weather Prediction (NWP) relies on fluid dynamics and complex physical principles. It requires **supercomputers** and extreme expertise in atmospheric physics.

The ML Paradigm

Machine Learning models learn patterns from **historical data** without solving physical equations. This offers a computationally efficient alternative for local forecasting.

Core Research Focus

Multi-Horizon

Examining performance degradation across 3-day, 5-day, and 7-day future time frames.

Multi-Output

Simultaneously predicting both Daily Maximum and Minimum air temperatures.

Comparison

Evaluating Linear Regression, Random Forest, SVR, and Multi-Layer Perceptrons.

Methodological Framework: CRISP-DM

- **Data Understanding:** Analysis of meteorological patterns in Cesena, Italy (2015-2025).
- **Data Preparation:** Aggregation of 6-hour measurements into daily max/min features.
- **Modeling:** Implementation of four regression architectures with hyperparameter tuning.
- **Evaluation:** Testing on unseen 2023-2025 data using RMSE, MAE, and R^2 metrics.

Study Area: Cesena, Italy

Coordinates: 44.18° N, 12.21° E

Period: 2015 – 2025

Source: Open-Meteo Platform

The dataset includes 16,072 observations of variables like humidity, pressure, cloud cover, and wind speed, aggregated into a 4,018-sample daily dataset.

Feature Engineering & Reduction

Multicollinearity Removal

Highly collinear features (>0.9 correlation) were removed to prevent instability:

- Temp vs Apparent Temp (0.991)
- Surface Pressure vs MSL Pressure (1.000)
- Soil Temp vs Apparent Temp (0.980)

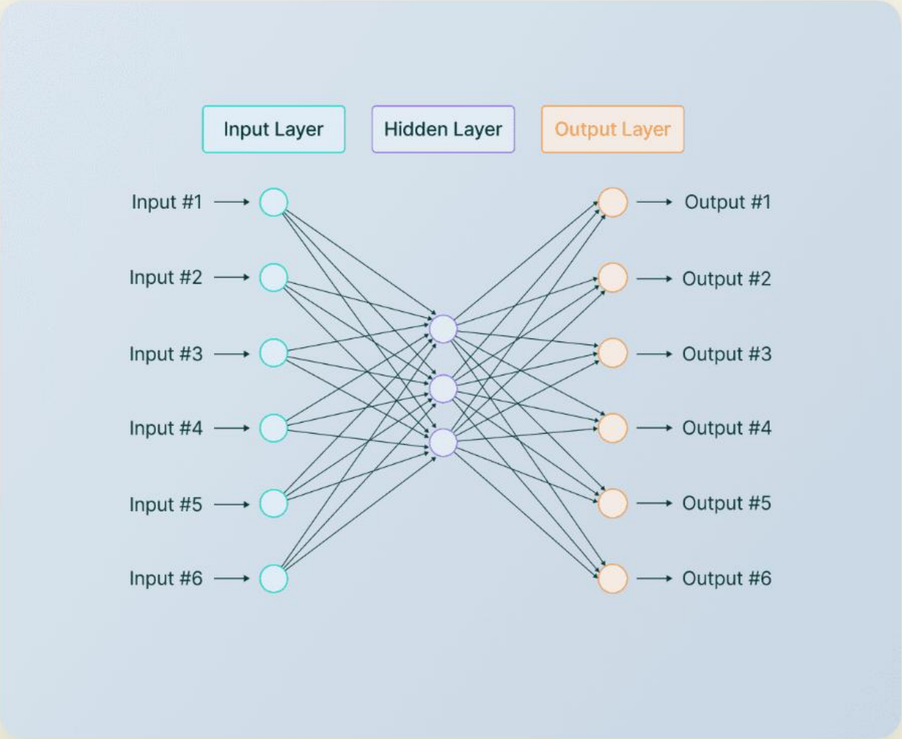
Temporal Decomposition

DateTime stamps were split into discrete numerical features to capture seasonality:

Hour, Month, Year, Day of Week, Day of Month.

Standardization: All features centered at zero mean with unit variance for stable convergence.

Evaluated Architectures



- **Linear Regression:** Statistical baseline model.
- **Random Forest:** Optimized with 200 trees and max depth of 20 via Grid Search.
- **SVR:** RBF kernel with C=100 and Gamma=0.1.
- **MLP (Neural Network):** (100, 50) hidden layer config, ReLU activation, L2 regularization (0.001).

Criteria for Success

Root Mean Square Error

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Coefficient of Determination

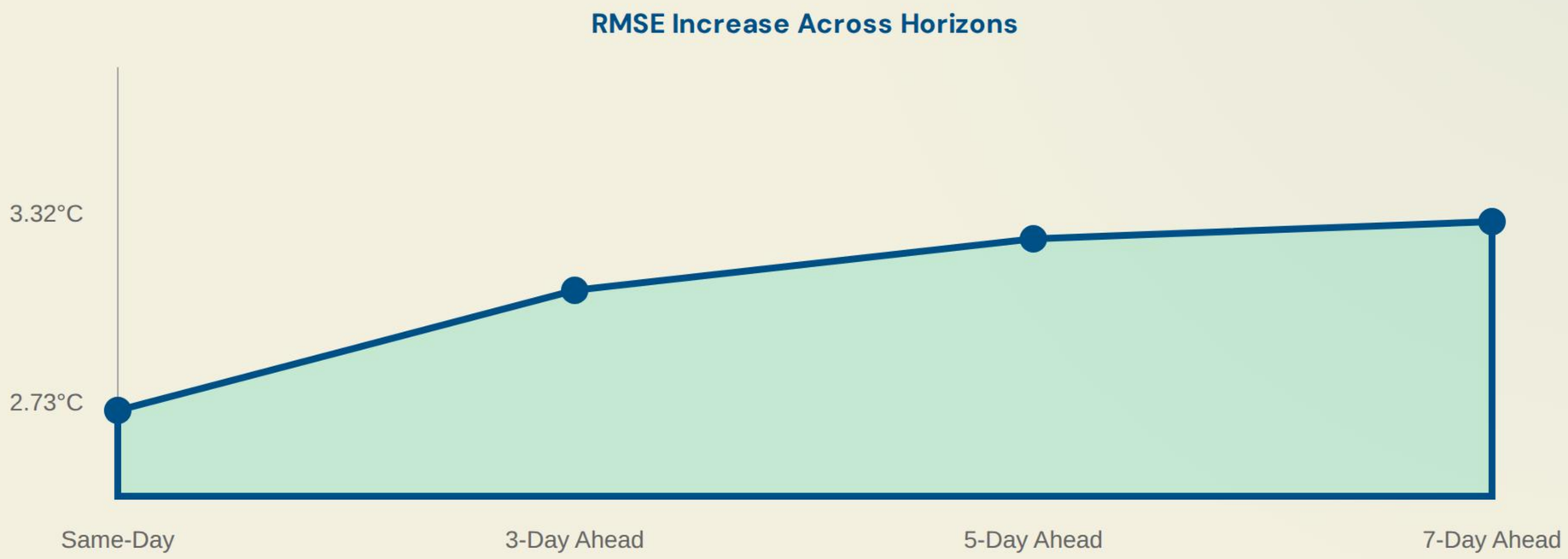
$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Metrics are averaged across both Maximum and Minimum temperature outputs.

Same-Day Prediction Performance

MLP achieved a 47% reduction in error compared to the linear baseline.

Multi-Step Horizon Forecasting (MLP)



Error increase diminishes at longer horizons as models converge toward climatological means.

Key Conclusions

87%

VARIANCE EXPLAINED (R^2)

3.3

MAX RMSE FOR 5-DAY

- **Non-Linear Superiority:** Neural networks are exceptionally well-suited for correlated multi-output tasks.
- **Reliability Window:** Accurate forecasting is highly viable within a 3-5 day window (RMSE < 3.3°C).
- **Efficiency:** Batch learning with structured features avoids the computational complexity of recurrent models.

Thank You

Questions & Discussion

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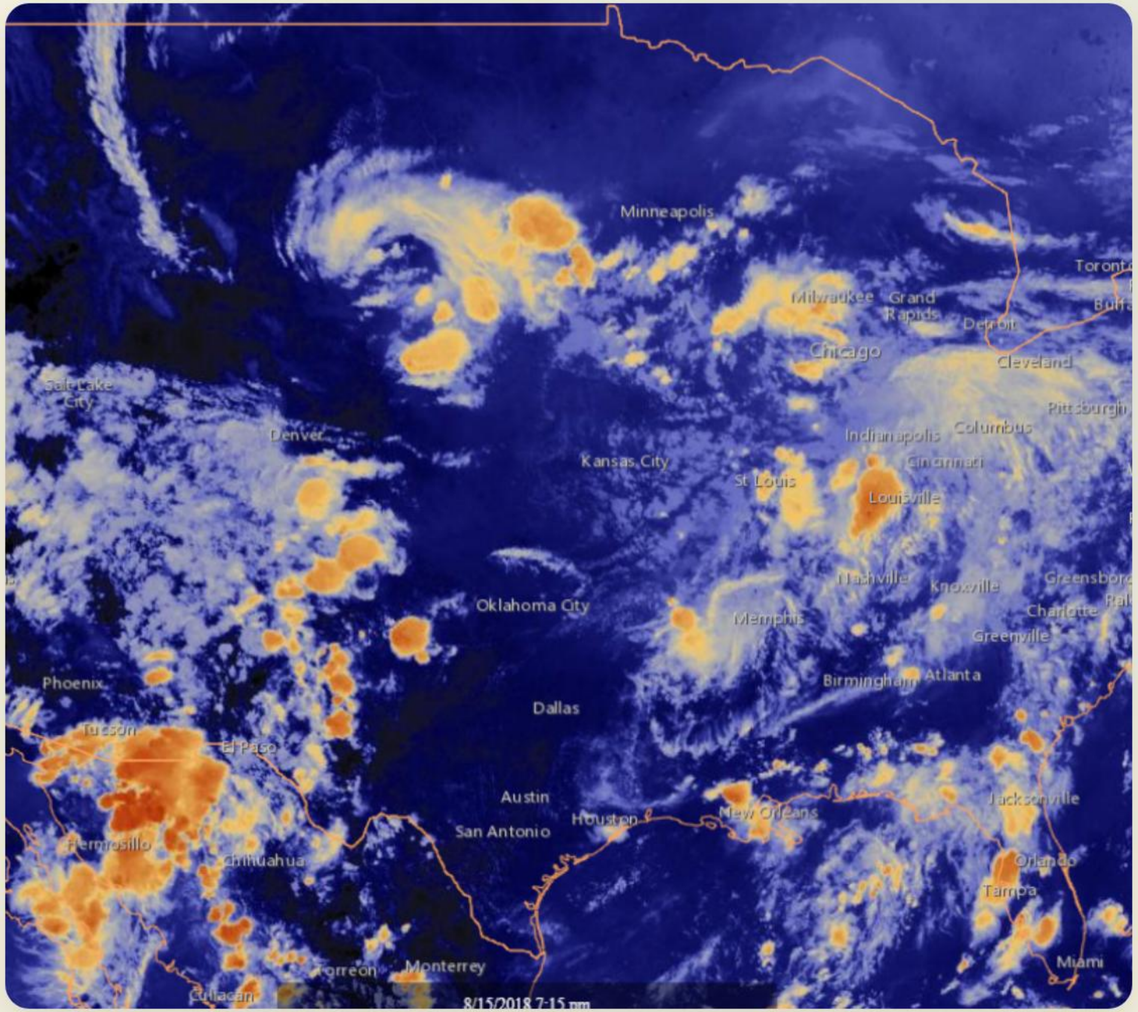

Image Sources

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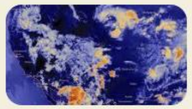

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